NAS Microfinance Management System (TZS)

Complete Product Blueprint (TZS Currency + Mandatory Client Referral Info), Financial & Overdue Engine, Appearance Mockups, Desktop/Mobile Key-Press Matrix & Implementation Plan

Property	Specification
Document Version	1.1.0 (October 20, 2026) — TZS Currency & Mandatory Referral Info
Primary Currency	TZS (Tanzanian Shilling) — Supports M-Pesa, Tigo Pesa, Airtel Money, HaloPesa, Bank, Cash
Client Referral Policy	Mandatory Referral / Guarantor on Every Client (Name, Phone, Relationship, Location/NIDA)
Target Platforms	Windows/Mac Desktop PC & Android/iOS Mobile Phone (Installable PWA + Cloud Sync Anywhere)
Multi-User Licensing	One-Time Activation Key System (Expires Immediately After Account Creation) + Super Admin Panel

1. Executive Summary & Vision (TZS)

A cross-platform Progressive Web App (PWA) for managing financial assistance (loans) in Tanzanian Shillings (TZS) with mandatory Client Referral / Guarantor tracking, automatic interest calculation, strict deadline & overdue rollover enforcement, real-time PC-to-Phone synchronization outside the local network, and a one-time activation key licensing system for multi-user expansion.

• All financial assistance, interest, penalties, and repayments tracked and formatted in Tanzanian Shillings (TZS)

• Mandatory Referral / Guarantor Information on every client (Referral Name, Phone Number, Relationship, Location/NIDA, and Referral Trust Chain analytics)

• Real-time cloud synchronization between Desktop PC and Mobile Phone over any network (Wi-Fi, 4G, 5G) with local IndexedDB offline fallback

• Instant dashboard visibility of Total Principal Offered (TZS), Interest %, Expected Profit (TZS), Total Expected Collection (TZS), Realized Profit (TZS), and Overdue Loans

• Automated time-bound interest & 'Or Else' overdue penalty/rollover engine (e.g., 30 days for 20% interest, plus automatic rollover or daily penalty after day 30)

• Multi-tenant SaaS architecture with single-use expiring Activation Keys and Super Admin management portal

2. Mandatory Client Referral / Guarantor Specification

Field	Requirement	Description & Purpose
referral_name	string (required)	Full name of the person who referred or guaranteed the client (e.g., Musa Juma)
referral_phone	string (required)	Phone number of the referral (+255 7XX XXX XXX) for 1-click Call or WhatsApp escalation
referral_relationship	enum/string (required)	How the referral knows or guarantees the borrower
referral_location_or_id	string (optional)	Physical shop/office address, NIDA ID, or notes about the referral
referred_by_borrower_id	UUID (optional)	Links to an existing borrower record if the referral is also an active/past client

3. Financial & 'Or Else' Overdue Engine in TZS

Mode	Formula	TZS Example (TZS 500,000 @ 20% for 30 Days)
SIMPLE_ROLLOVER Fixed Principal Cycle Rollover (Default)	F_penalty = ceil(days_overdue / D) * (P * r)	Loan of TZS 500,000 @ 20% for 30 days (TZS 600,000 due). On Day 31, Cycle 2 starts and adds +TZS 100,000. New Total Expected = TZS 700,000.

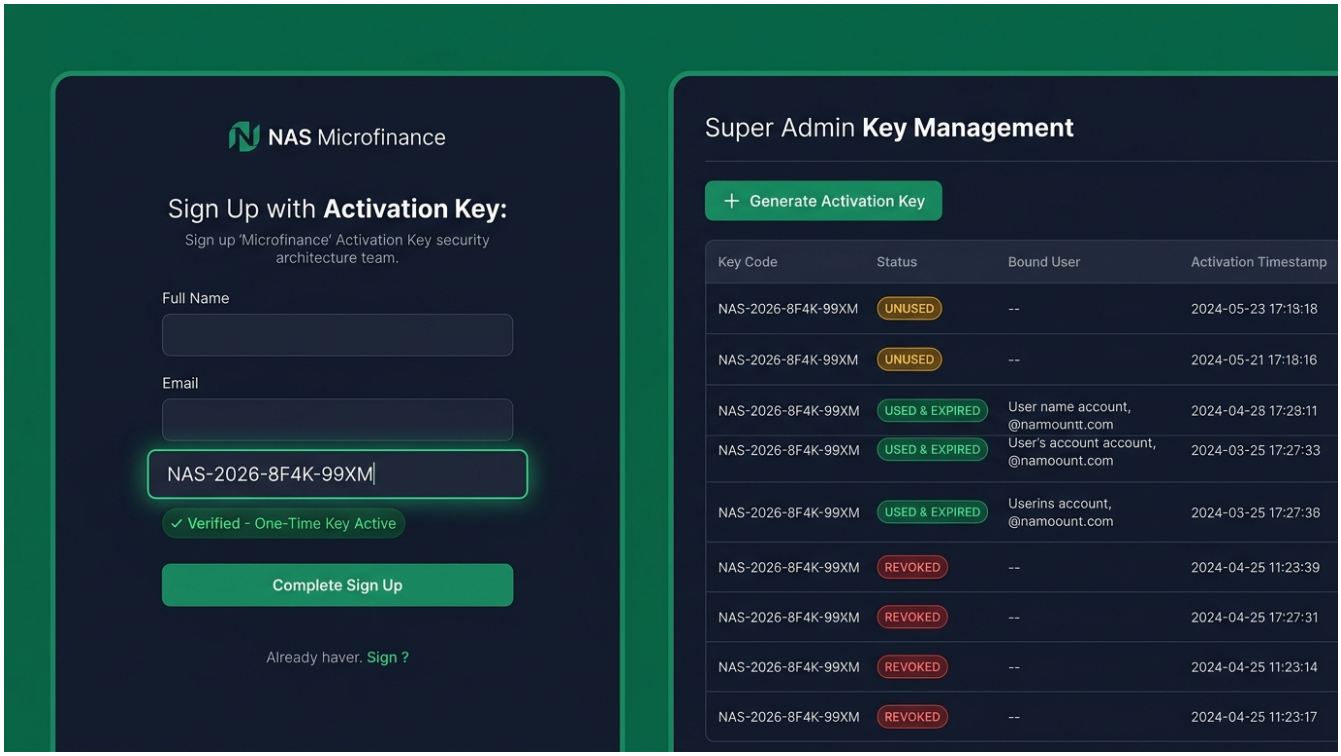
COMPOUND_ROLLOVER Compound Unpaid Balance Rollover	New_Balance = B_rem * ((1 + r) ^ ceil(days_overdue / D))	Loan of TZS 500,000 @ 20% for 30 days (TZS 600,000 due). On Day 31, adds 20% of TZS 600,000 (+TZS 120,000). New Total Expected = TZS 720,000.
DAILY_PENALTY Daily Late Percentage Fee	F_penalty = P * r_daily * days_overdue	Loan of TZS 500,000 with 1% daily late fee. 5 days overdue adds +TZS 25,000. New Total Expected = TZS 625,000.
MANUAL_PENALTY Manual Alert & Custom TZS Penalty	F_penalty = manual_tzs_amount_entered_by_lender	Lender clicks '+ Apply Penalty' or '+ Extend 7 Days' from the Loan Detail drawer.

4. Screen-by-Screen Appearance Mockups & Interaction Matrix

SCREEN_01: Login & Activation Key Onboarding (TZS Workspace)

Desktop Layout: Centered split-tab card (520px width) on dark slate-navy canvas with live sync indicator in top bar.

Mobile Layout: Full-screen vertical card with sticky bottom submit button and native paste button for activation key.

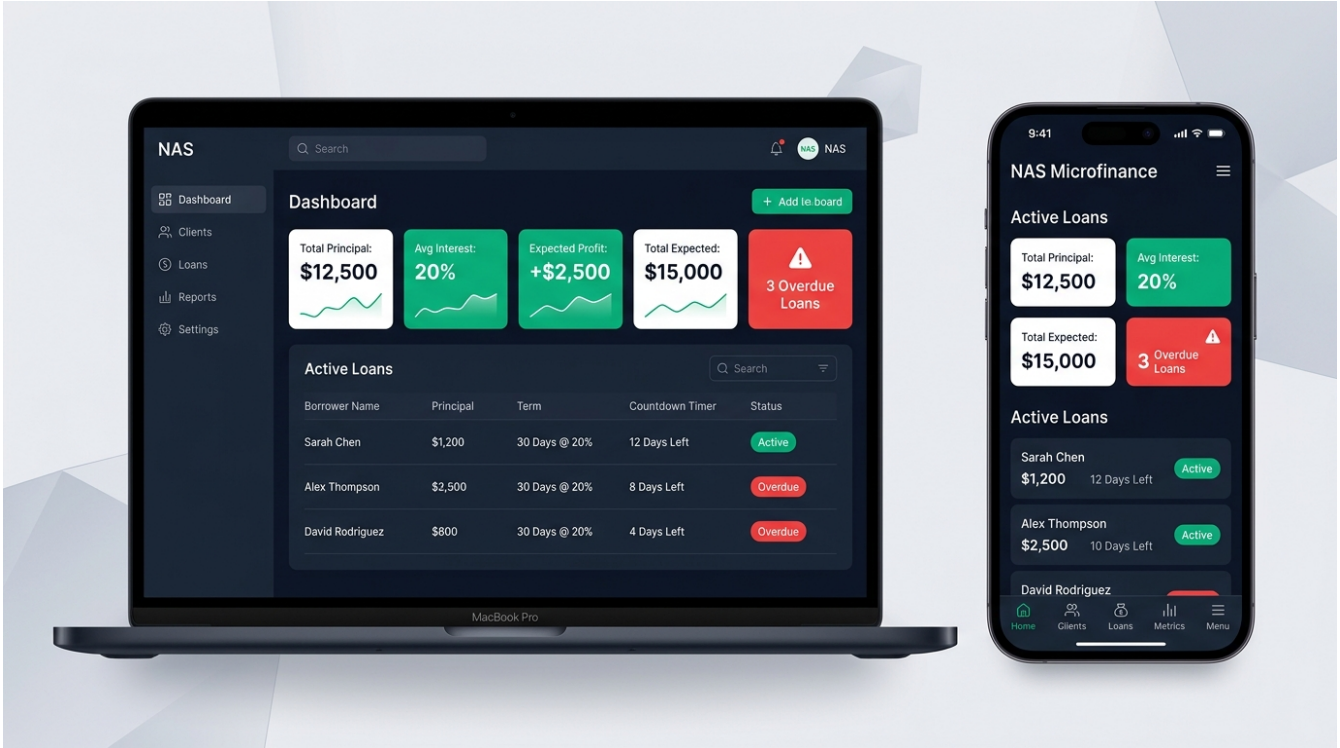


Feature / Step	Desktop Key / Click	Mobile Touch	System Response
Switch to Activation Tab	Press 'Alt+2' or click '[Activate New Account]' tab	Tap '[Activate New Account]' segmented control	Reveals Step 1 (Activation Key Input) and auto-focuses the key field.
Enter / Paste Activation Key	Type or press 'Ctrl+V' to paste 16-char key (e.g., NAS-2026-8F4K-99XM)	Tap '[Paste Key]' chip or type on uppercase alphanumeric keyboard	Auto-inserts hyphens every 4 chars, verifies key status against cloud DB in <150ms, and shows green 'Verified Single-Use Key' badge.
Complete Profile & Submit	Press 'Tab' through Business Name, Phone (+255...), Password -> Press 'Enter'	Fill fields using mobile 'Next' key -> Tap '[Activate & Launch TZS Workspace]'	Burns activation key atomically, initializes isolated TZS workspace, and redirects to Executive Dashboard.

SCREEN_02: Executive Dashboard in TZS (Desktop & Mobile)

Desktop Layout: Left navigation sidebar (240px) + Top Command Bar + 4 TZS KPI Summary Cards (Principal Offered: TZS 12,500,000, Expected Profit: +TZS 2,500,000, Total Expected Collection: TZS 15,000,000, Overdue Alert: TZS 1,800,000) + Active Loans & Referrals Table.

Mobile Layout: Top status header + Hero Expected Collection Card (TZS 15,000,000) + 2x2 KPI Grid + Filter Chips + Swipeable Loan Cards showing Client & Referral + Floating '+' Button + Bottom Nav Bar.

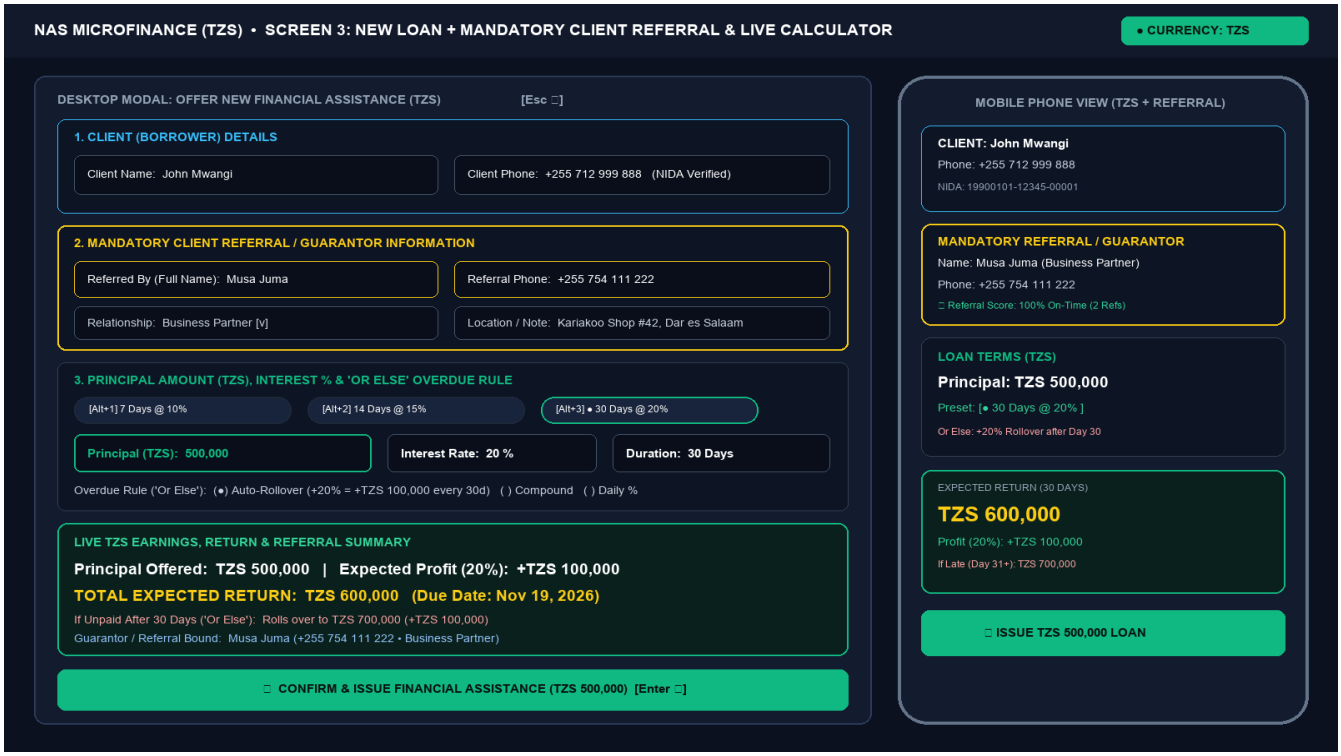


Feature / Step	Desktop Key / Click	Mobile Touch	System Response
Global Search by Client or Referral	Press 'Ctrl+K' or '/' -> type Client or Referral name/phone	Tap Search icon in top bar -> type Client or Referral name/phone	Instantly filters borrowers and loans by either the borrower's details or their Referral/Guarantor's details.
Launch New TZS Loan Form	Press 'N' or click '[+ New Loan]' button	Tap floating emerald '(+)' button at bottom-right	Opens Screen 3 (New Financial Assistance + Mandatory Referral Modal / Bottom Sheet).
Filter Overdue Loans	Press 'O' or click red 'OVERDUE' KPI card	Tap '[Overdue]' filter pill	Filters list to loans past their due date, sorted by urgency and TZS rollover amount.
Quick Row Navigation & Referral Escalation	Press 'J'/'K' to select loan -> 'Enter' for details, 'P' to record payment, 'W' to message Client, or 'R' to contact Referral	Tap loan card for details, Swipe Right to record payment, Tap '[Client ■]' or Tap '[Ref ■]'	Opens Loan Detail Drawer, Repayment dialog, or pre-filled WhatsApp/Call to Client or Referral.

SCREEN_03: New Financial Assistance (Client + Referral Info & Live TZS Calculator)

Desktop Layout: 2-column modal dialog: Left column for (1) Client Details, (2) Mandatory Referral/Guarantor Details, (3) Principal in TZS, Rate %, Duration, and Overdue Rule; Right column for Live TZS Earnings & Referral Summary Card.

Mobile Layout: Slide-up bottom sheet with Client & Referral inputs, Quick Preset pills ('7d @ 10%', '14d @ 15%', '30d @ 20%'), decimal keypad inputs, and sticky live TZS preview banner.

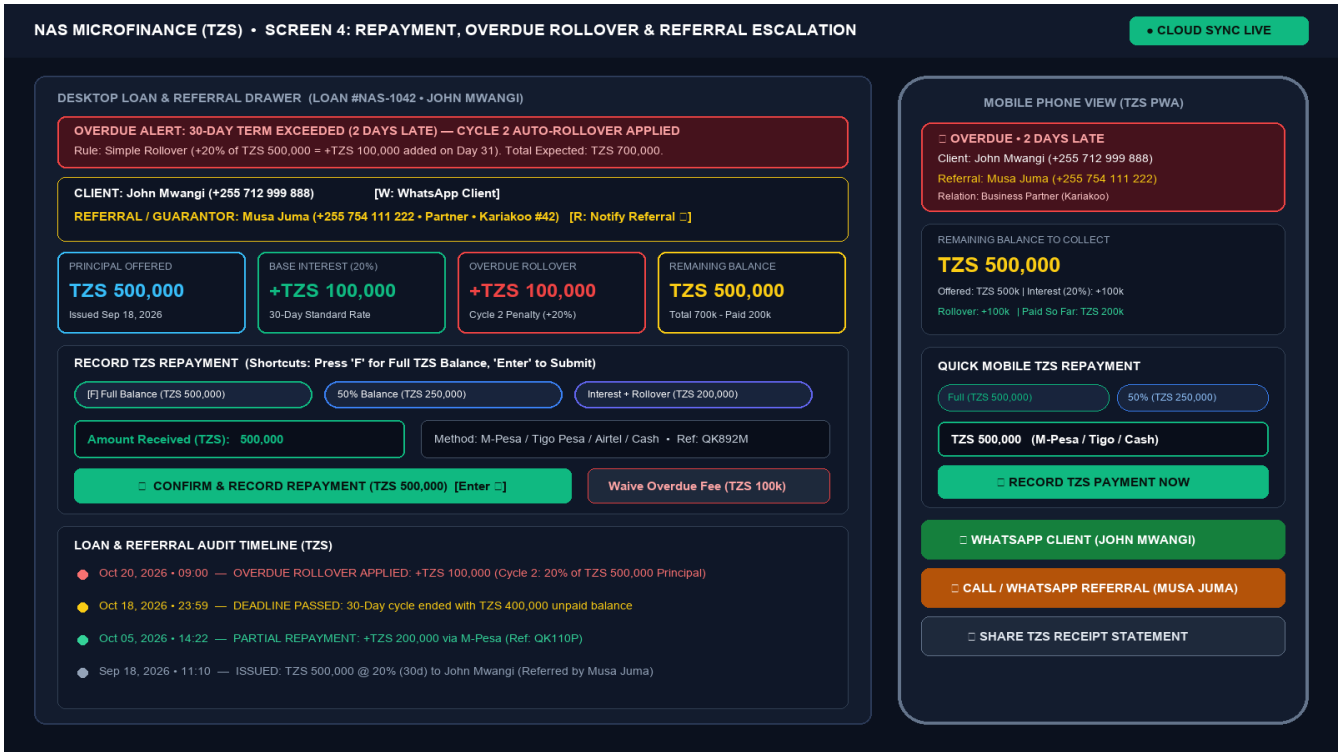


Feature / Step	Desktop Key / Click	Mobile Touch	System Response
Enter Client Information	Type Client Name & Phone (+255...) or select existing client from dropdown	Tap Client input -> pick existing client chip or enter new name & phone	If existing client is selected, auto-fills their saved Referral Information and displays their repayment trust score.
Enter Mandatory Referral / Guarantor Information	Press 'Tab' -> Enter Referral Full Name, Referral Phone (+255...), Relationship (e.g., Business Partner), and Location/NIDA note	Tap Referral fields (or tap '[Pick Existing Client]' as Referee)	Validates referral fields are complete and checks if this referral has any other referred clients currently Overdue.
Enter Principal Offered in TZS	Press 'Tab' -> Type numeric TZS amount (e.g., '500000')	Tap Principal (TZS) field -> numeric keypad opens -> tap '500000'	Auto-formats as 'TZS 500,000' and live preview immediately computes Expected Interest (+TZS 100,000) and Total Return (TZS 600,000).
Set Rate %, Duration Days & Overdue Rule	Press 'Alt+1' (7d@10%), 'Alt+2' (14d@15%), 'Alt+3' (30d@20%), and choose Overdue Rule	Tap preset pill '[30d @ 20%]' and tap Overdue Rule card	Displays Due Date and exact Day 31+ overdue projection in TZS (e.g., 'If unpaid after 30 days: becomes TZS 700,000').
Issue TZS Financial Assistance	Press 'Enter' or 'Ctrl+Enter'	Tap '[Confirm & Issue TZS Loan]'	Saves client, referral, and loan to DB, updates TZS dashboard KPIs, syncs across PC & Phone, and offers 1-click Receipt sharing.

SCREEN_04: Loan Detail, Overdue 'Or Else' Rollover, Referral Escalation & Repayment Drawer

Desktop Layout: Slide-over right drawer (580px) showing Overdue Banner, Client & Referral Contact Card, TZS Breakdown Cards, Mobile Money / Cash Repayment form, and Audit Timeline.

Mobile Layout: Full-screen detail view with 1-tap Client & Referral Call/WhatsApp buttons and sticky bottom '[Record TZS Payment]' bar.

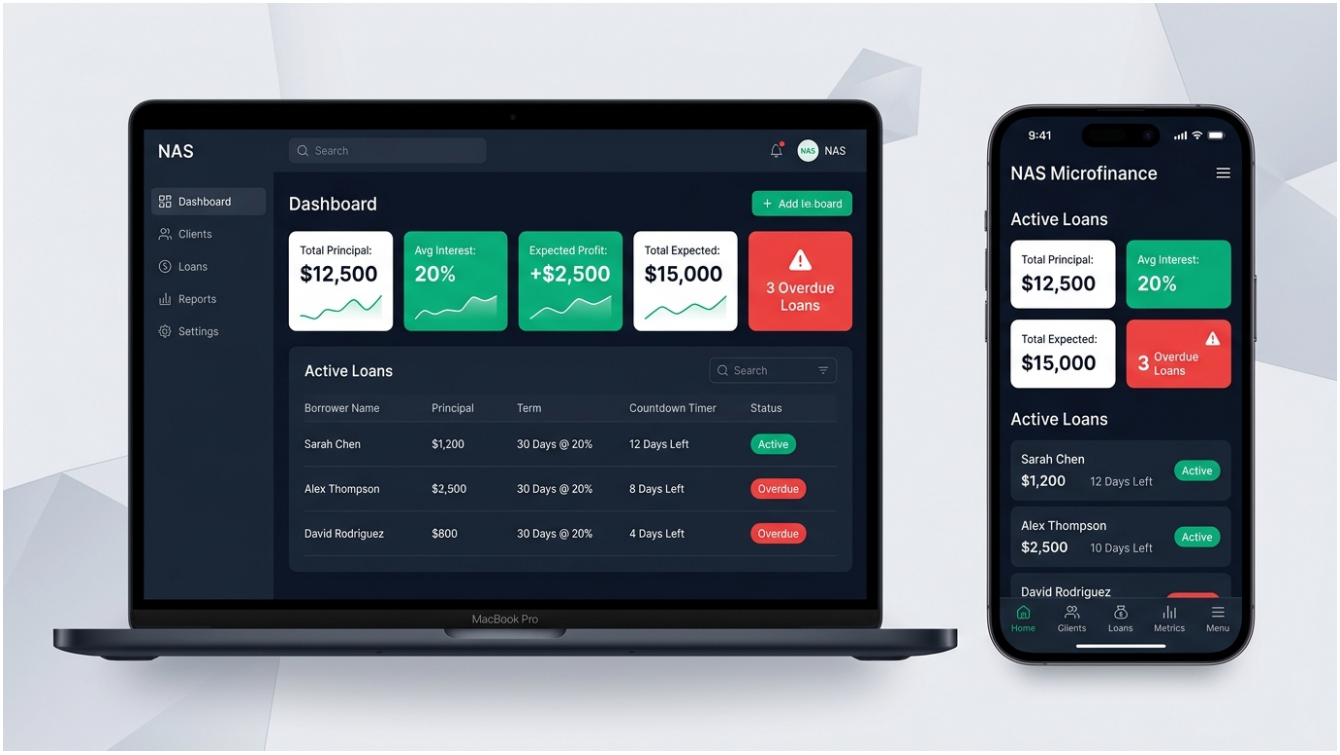


Feature / Step	Desktop Key / Click	Mobile Touch	System Response
One-Click Full TZS Settlement	Press 'F' (Full Balance) then 'Enter'	Tap '[Full Balance (TZS 500,000)]' chip then tap '[Record Payment]'	Fills exact remaining TZS balance, marks loan as PAID, and transfers interest to Realized Profit (TZS).
Partial Repayment via Mobile Money / Cash	Type partial TZS amount -> Select M-Pesa / Tigo Pesa / Airtel Money / Cash -> Press 'Enter'	Enter partial TZS amount -> Select payment method -> Tap '[Record Payment]'	Deducts payment from remaining TZS balance, updates progress bar, and logs transaction reference in audit history.
Notify Referral / Guarantor on Overdue Loan	Press 'R' or click '[Notify Referral ■]'	Tap '[Notify Referral ■]' or '[Call Referral ■]'	Opens WhatsApp or Phone Dialer to the Referral's phone number with loan & overdue details.

SCREEN_05: Client & Referral Directory + Credit Risk Profile

Desktop Layout: Searchable data table with columns: Client Name & Phone, Referred By (Name, Phone, Relation), Active Loan (TZS), Lifetime Profit Earned (TZS), and Trust Score.

Mobile Layout: Contact cards displaying Client & Referral info with 1-tap Call, WhatsApp, and Trust Score badge.

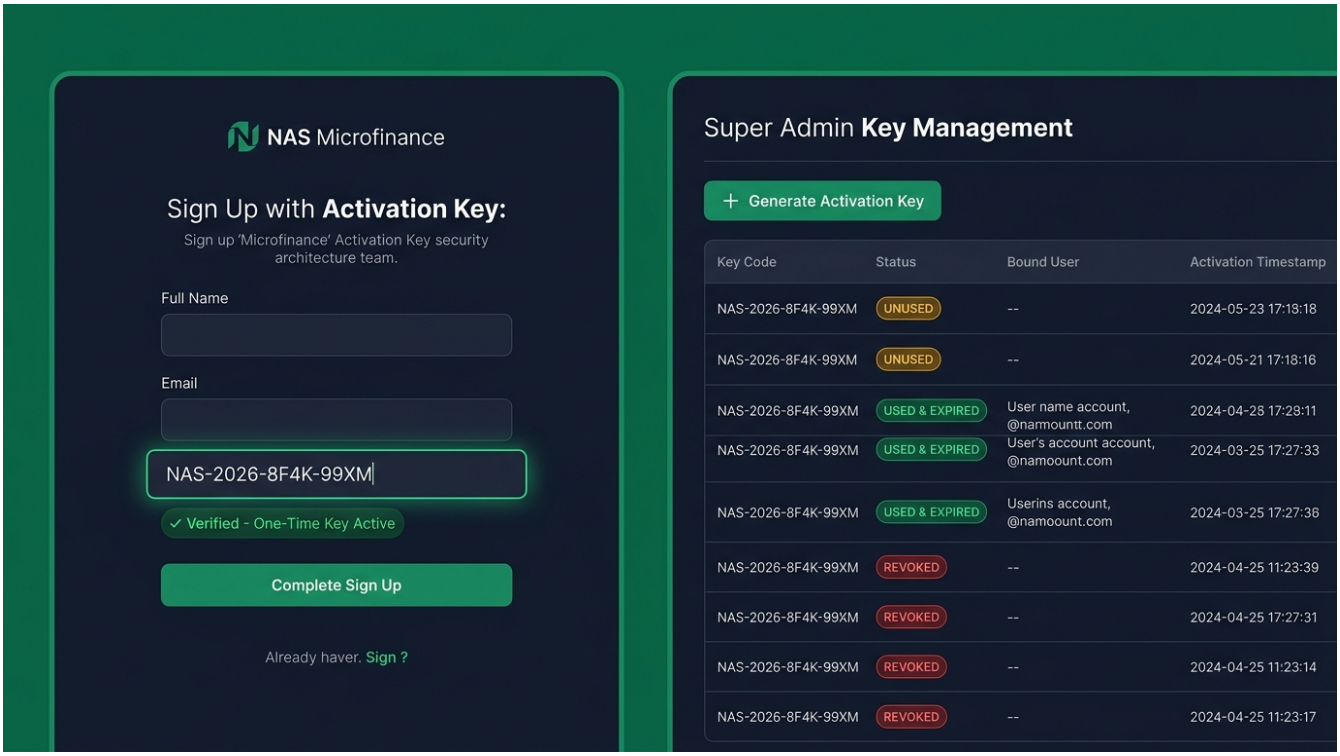


Feature / Step	Desktop Key / Click	Mobile Touch	System Response
Inspect Client & Referral Chain	Press 'B' -> Select client with 'Up/Down' -> Press 'Enter'	Tap 'People' tab -> Tap client card	Shows complete TZS borrowing history, guarantor/referral contact card, and any other clients referred by the same person.

SCREEN_06: Super Admin Activation Key Generator & User Manager

Desktop Layout: Admin console with Key Generator form at top and filterable Keys & Active Tenants table below.

Mobile Layout: Admin tab with 1-tap '[Generate Key]' card and shareable key list.



Feature / Step	Desktop Key / Click	Mobile Touch	System Response

Generate New Activation Key	Press 'K' -> Press 'G' or click '[Generate Key]'	Tap Admin Key icon -> Tap '[Generate Key]'	Generates unique key 'NAS-2026-XXXX-XXXX' with status UNUSED.
Copy / Share Key with New User	Press 'C' or click '[Copy Key]'	Tap '[Share via WhatsApp]'	Copies key + invite instructions so the recipient can create their TZS lender account before the key expires.

5. Formal Step-by-Step Implementation Blueprint

Phase 1: Project Scaffolding & PWA Setup

• Next.js TypeScript + Tailwind CSS project initialized in workspace

• PWA manifest.json, icons, and offline service worker configured for PC & Mobile installation

• Responsive layout shell (Desktop Sidebar + Mobile Bottom Navigation)

Phase 2: TZS Financial Engine & Client Referral Model

• Pure TypeScript financial module (src/lib/finance-engine.ts) with TZS currency formatting

• Mandatory Client Referral / Guarantor validation and referral-chain risk badges

• Support for Principal (TZS), Interest %, Duration Days, Due Date, and all 4 Overdue 'Or Else' modes

Phase 3: Multi-User Auth & One-Time Activation Key System

• Database schema & Row-Level Security (RLS) for strict per-user isolation

• Sign-Up with Activation Key verification & atomic key burn (UNUSED -> USED)

• Super Admin Key Generator & User Licensing management screen

Phase 4: Executive TZS Dashboard, Loan + Referral Modal & Repayment Drawer

• Executive Dashboard with live TZS KPI cards (Offered, Interest %, Expected Profit, Overdue)

• New Financial Assistance modal/sheet with Client + Mandatory Referral fields and live TZS calculator

• Loan Detail & Repayment drawer with Mobile Money methods (M-Pesa, Tigo Pesa, Airtel Money, HaloPesa, Cash), 1-click Full Balance settlement, and Client/Referral WhatsApp reminders

Phase 5: Cross-Network Cloud Sync, Offline Queue & Keyboard Shortcuts

• Real-time sync via Supabase / Cloud API + local IndexedDB persistence

• Global desktop keyboard shortcuts (N, P, R, W, D, L, B, K, O, Ctrl+K, Esc) and mobile touch gestures

• Shareable TZS PDF/PNG Receipt generator showing Client, Referral, and Loan breakdown

Phase 6: Stage 2 Production Hardening & Live Operations (v1.2.0)

• Upgrade 1: Front-door authentication lock + in-app Admin/Lender Email, Phone & Password changer

• Upgrade 2: One-click 'Clear Sample Data (Start Fresh at TZS 0)' portfolio reset

• Upgrade 3: Full Edit & Delete controls for Clients, Mandatory Referrals, and TZS Loans

• Upgrade 4: Branded 192x192 & 512x512 PWA Icons, Service Worker, and in-app 'Install App' button

• Upgrade 5: One-click 'Export TZS Portfolio to Excel (CSV)' backup generator